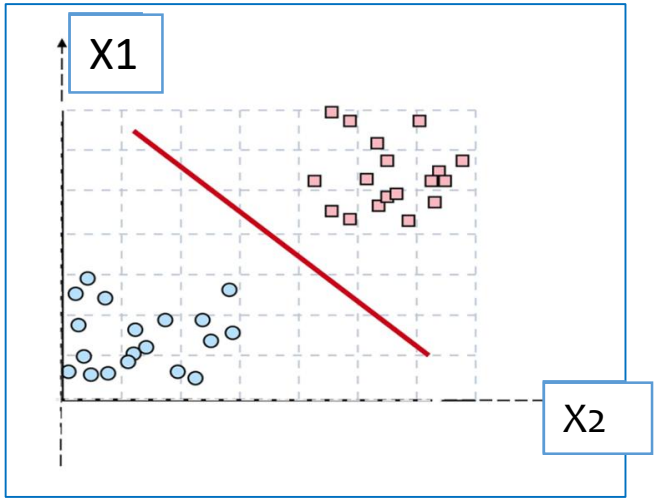
Equation of Straight Line

$ax + by + c = 0$



Equation of Straight Line

$ax + by + c = 0$

$$ax_1 + bx_2 + c = 0$$

If $x = x_1$
$y = x_2$

If $a = w_1$
$b = w_2$

$$w_1x_1 + w_2x_2 + c = 0$$

$$w_1x_1 + w_2x_2 = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} [x_1, x_2,]$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} [x_1, x_2,] = w^T x$$

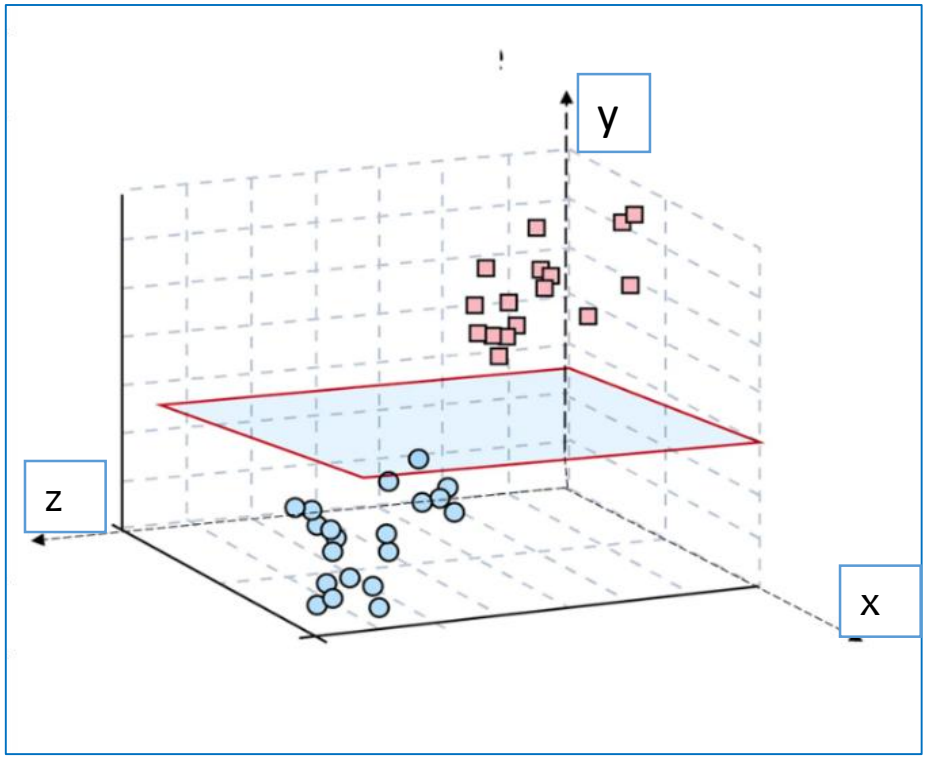
$$w^T x + c = 0$$

Equation of Straight Line 3 Dimension

$$ax + by + cz + c = 0$$

$$ax_1 + bx_2 + cx_3 + c = 0$$

```
2x + 3y + 4z + 5 =0
X – 2y + 3z – 6 =0
x- 2y + 3z – 6 = 0
```



Equation of Straight Line

$ax + by + cz + c = 0$

$$ax_1 + bx_2 + cx_3 + c = 0$$

$$w_1x_1 + w_2x_2 + w_3x_3 + c = 0$$

If $a = w_1$

$b = w_2$

$c = w_3$

$$w_1x_1 + w_2x_2 + w_3x_3 + c = 0$$

$$w_1x_1 + w_2x_2 + w_3x_3 = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} [x_1, x_2, x_3]$$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} [x_1, x_2, x_3] = w^T x$$

$$w^T x + c = 0$$

4D and Beyond

$$w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 \ldots + w_nx_n + c = 0$$

$$w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 \ldots\ldots + w_nx_n = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ . \\ w_n \end{bmatrix} [x_1, x_2, x_3, x_4 \ldots x_n]$$

$$w^T x + c = 0$$

$$w_1x_1 + w_2x_2 + w_3x_3 \ + w_4x_4 \ldots \ + w_nx_n + \ c \ = 0$$

Can I denote the same as below as well

$$w_1v_1 + w_2v_2 + w_3v_3 \ + w_4v_4 \ldots \ + w_nv_n + \ c \ = 0$$

$$w^T v + \ c \ = 0$$

$y = m\,x + c$

$-1x + 2y - 2 = 1$

$-1x + 2y - 2 = 1$

$-1x + 2y = 1 + 2$

$-1x + 2y = 3$

$2y = 1x + 3$

$y = \frac{1}{2}\,x + \frac{3}{2}$

$y = 0.5\,x + 1.5$

$-1x + 2y - 3 = 0$

$-1x + 2y - 3 = 0$

$-1x + 2y = +3$

$2y = 1x + 3$

$y = \frac{1}{2}\,x + \frac{3}{2}$

$y = 0.5\,x + 1.5$

Equation of Straight Line

$ax + by + c = 0$

$by + c = -ax$

$by = -ax + (-c)$

$y = \frac{-a}{b}x + \frac{-c}{b}$

Equation of Straight Line

y = **m** x + **c**

m OR $\frac{-a}{b}$ = (Slope or Coefficient) slope of the line

c OR $\frac{-c}{b}$ = Intercept (where the best-fit line meets the Y axis).

$$2x+3y-6=0$$

$$4x-y+8=0$$

$$3x+2y+7=0$$

$$5x-2y+1=0$$

$$-1x+2y-2=1$$